Privacy and Predictive Utility in Federated Learning for Hospitals

A Reproducible Study Using the Diabetes 130-US Hospitals Dataset

October 13, 2026

Abstract

This project studies whether a small hospital benefits from federated learning (FL), and how record-level differential privacy (DP) changes predictive utility and vulnerability to a loss-based membership inference attack (MIA). We use the Diabetes 130-US Hospitals dataset and simulate ten clients: one small hospital, one large hospital, and eight background hospitals. The main model is logistic regression trained for five rounds using Federated Averaging. Private runs use client-side DP-SGD with target privacy budgets $\varepsilon \in \{1, 2, 4, 8, 16\}$ and $\delta = 10^{-5}$. Results are averaged over five random seeds.

Without DP, the small hospital's global model achieved a holdout AUROC of 0.626 in the random partition and 0.639 in the age-shifted partition. This was substantially better than training a local logistic regression on only 400 records, and close to a pooled centralized baseline. Excluding the small hospital from training produced almost identical utility, because its 400 training records represented only 0.73% of all federated training records. DP runs produced lower utility, with no clear improvement as ε increased over the tested range. The loss-based MIA was near chance even without DP, so the experiment did not provide evidence that DP reduced an already measurable attack advantage. The study therefore demonstrates a utility benefit from receiving a non-private global model, but it cannot identify a privacy budget at which MIA risk disappears.

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1 The project in plain language

Imagine ten hospitals that want one readmission-prediction model. A large hospital has many patient records, while a small hospital has only a few hundred. If all records could be placed in one database, training would be straightforward, but sharing raw medical records creates privacy and governance concerns.

Federated learning keeps each hospital's records in its own local partition. Every hospital receives the same model, trains it locally, and returns model parameters rather than patient rows. A server averages the returned parameters to form a new global model. This reduces direct data sharing, but it does not by itself prove privacy: model updates and trained models can still reveal information.

Differential privacy addresses this second problem by limiting how much one patient can influence training. In this project, each client's per-patient gradients are clipped and Gaussian noise is added before the local update is returned. More noise usually strengthens privacy but can damage prediction quality.

The project asks whether the small hospital should:

1. train only on its own data;
2. receive a global model without contributing its own data;
3. contribute without differential privacy; or
4. contribute while all participating clients use DP-SGD.

> **In plain language.** The clearest result is that the small hospital benefits from receiving the non-private global model. Its own contribution changes that model very little because it supplies only 400 of 55,167 training records. The private models perform worse in the tested five-round schedule. The chosen attack is already at chance without DP, so it cannot demonstrate an additional attack reduction caused by DP.

2 Background concepts

2.1 Binary prediction and logistic regression

The target is binary:

$$y_i = \begin{cases} 1, & \text{patient } i \text{ is readmitted within 30 days,} \\ 0, & \text{otherwise.} \end{cases}$$

Logistic regression forms a linear score

$$z_i = \mathbf{w}^\top \mathbf{x}_i + b$$

and converts it to a probability using the sigmoid function

$$\hat{p}_i = \frac{1}{1 + e^{-z_i}}.$$

The model is trained with binary cross-entropy loss:

$$\ell_i = -y_i \log(\hat{p}_i) - (1 - y_i) \log(1 - \hat{p}_i).$$

Low loss means that the model assigned a high probability to the correct outcome. Logistic regression was appropriate as a simple baseline and has only 136 trainable parameters for 135 input features: 135 weights and one bias.

2.2 Federated learning

Federated learning trains a shared model while raw records remain in local client datasets [3]. In round t:

1. the server sends global parameters $\mathbf{w}_t$ to every client;
2. each client trains a local copy on its own training records;
3. each client returns its updated parameters and training-set size; and
4. the server computes a sample-size-weighted average.

For client k with n_k training records, Federated Averaging uses

$$\mathbf{w}_{t+1} = \sum_{k=1}^{K} \frac{n_k}{\sum_j n_j} \mathbf{w}_{t+1}^{(k)}.$$

This weighting matters greatly in this project. Client 0 has only 400 training records and therefore contributes approximately

$$\frac{400}{55,167} = 0.00725 = 0.73\%$$

of a normal aggregated update. Consequently, including or excluding that client should have only a small numerical effect on the global parameters.

2.3 IID and non-IID data

IID means that clients are treated as samples from roughly the same underlying population. Non-IID means their distributions differ. Real hospitals are often non-IID because they serve different ages, regions, disease severities, and referral patterns.

This study creates a controlled non-IID setting by making 80% of the small hospital's selected population aged 70 years or above. This changes the small hospital's case mix. It does not create ten different age-restricted hospitals; only client 0 and its holdout pool are deliberately age-shifted.

2.4 Differential privacy

An algorithm is approximately (ε, δ)-differentially private if its output distribution changes only within a bounded amount when one person's record is added or removed [4]. Informally:

- smaller ε means a stronger privacy guarantee;
- larger ε allows a greater influence from one record;
- δ is a small failure-probability term; and
- privacy is a property of the randomized training procedure, not a statement that an observed attack must have AUROC exactly 0.5.

The implementation uses DP-SGD [5]. For each batch it conceptually performs three steps:

1. calculate a separate gradient g_i for each patient;
2. clip each gradient to maximum norm C,

$$\bar{g}_i = g_i \min\left(1, \frac{C}{\|g_i\|_2}\right);$$

3. aggregate the clipped gradients and add Gaussian noise proportional to σC.

Clipping limits one patient's influence. Noise makes it harder to infer whether that patient participated. Both operations can reduce model utility. The Opacus library performs per-sample gradient calculation, Poisson sampling, clipping, noise addition, and privacy accounting [9].

2.5 Membership inference

A membership inference attack asks: "Was this person used to train the model?" [6, 7]. The loss-based attack used here relies on the idea that an overfitted model may assign lower loss to training members than to unseen non-members.

For every attacked patient, the attack score is

$$s_i = -\ell_i.$$

The minus sign is used because lower prediction loss should produce a higher membership score. Attack AUROC measures how well this continuous score ranks members above non-members. An AUROC of 0.5 means no ranking advantage; values above 0.5 indicate increasing attack success. Values slightly below 0.5 do not imply "better than perfect privacy"; they normally indicate sampling variation or a reversed weak signal.

Attack accuracy uses a fixed threshold equal to the members' mean loss. Because this threshold is calibrated using known member losses, AUROC is the cleaner primary metric in this study.

3 Research questions

The analysis is organized around four questions:

RQ1.
At what target privacy budget ε does the membership inference attack on the small hospital fall to chance?

RQ2.
Does the small hospital benefit from contributing to federated training rather than only receiving the global model?

RQ3.
Does training only on the small hospital's local data outperform participation in, or receipt of, the federation?

RQ4.
Does the same target ε lead to different observed privacy and utility outcomes for the small and large hospitals?

These questions need careful interpretation. In particular, RQ1 can identify a transition only if the attack is detectably above chance without DP. If the non-private attack is already at chance, the experiment has no measurable attack advantage for DP to remove.

4 Data and preprocessing

4.1 Source data

The Diabetes 130-US Hospitals for Years 1999–2008 dataset contains 101,766 inpatient encounters collected from 130 US hospitals and integrated delivery networks [1, 2]. Patients have diabetes, stayed in hospital for between one and fourteen days, underwent laboratory testing, and received medication. The raw outcome has three categories: readmitted within 30 days, readmitted after 30 days, and no recorded readmission.

The project defines the positive class as readmission within 30 days. Both later readmission and no readmission are coded as class 0. After preprocessing, 6,275 of 69,963 patients are positive, a prevalence of 8.97%. This class imbalance is why accuracy alone would be misleading: a model predicting "not readmitted" for everyone would already be about 91% accurate.

4.2 Cohort construction

The cohort was constructed as follows:

Step	Records remaining
Raw encounters	101,766
Keep earliest encounter for each patient	71,518
Exclude terminal/hospice discharge dispositions	69,973
Remove records missing the primary diagnosis	69,963

Keeping one encounter per patient prevents the same person from appearing in both training and evaluation partitions. The code sorts by `encounter_id` and keeps the first encounter. This assumes encounter identifiers preserve chronological ordering. Terminal-care and hospice discharges are removed because subsequent readmission is not a meaningful or comparable outcome for these cases.

The final target is

$$\texttt{target} = 1(\texttt{readmitted} = \text{“<30”}).$$

Identifiers are then removed. They identify records but should not be treated as generalizable predictors.

4.3 Missingness analysis

Missingness was measured after patient deduplication and terminal/hospice exclusion, because this is the cohort on which later preprocessing operates.

Variable	Missing	Treatment
Weight	96.02%	Removed
Maximum glucose serum result	95.22%	“Not Tested” category
A1C result	81.64%	“Not Tested” category
Medical specialty	48.07%	Removed
Payer code	43.47%	Removed
Race	2.74%	Excluded from the feature set
Primary diagnosis	0.01% (10 rows)	Rows removed
Secondary diagnosis	0.42%	“No Diagnosis” category
Tertiary diagnosis	1.75%	“No Diagnosis” category

Weight, payer code, and medical specialty were removed because imputing variables with such extensive missingness would require strong assumptions. Laboratory missingness was handled differently: an absent glucose or A1C result can mean that the test was not performed, which is itself potentially informative. It was therefore represented explicitly as “Not Tested.” Only ten eligible records lacked a primary diagnosis, so removing them had negligible effect on cohort size. Missing secondary and tertiary diagnoses were retained as “No Diagnosis.”

Race and gender were excluded, but not because both were highly missing. This is a design choice that limits what can be concluded: the project cannot measure performance or privacy disparities by race or gender, and removing protected attributes does not ensure fairness because other variables may act as proxies.

4.4 Diagnosis representation

ICD-9 diagnosis codes are identifiers, not continuous quantities. Treating code 800 as numerically twice code 400 would be meaningless. One-hot encoding every exact code would preserve detail but create many rare indicators. The project instead maps each of the three diagnosis fields to broad ICD-9 disease chapters, such as circulatory, respiratory, digestive, and injury/poisoning. Diabetes codes (250.x) receive their own category because they are especially relevant to this cohort. V and E codes are placed in an external/supplemental category.

This is a bias–variance trade-off:

- **benefit:** fewer, better-supported indicators and easier interpretation;
- **cost:** loss of detail between diagnoses inside the same broad group.

The grouping is justified by the hierarchical meaning of ICD-9 ranges, not by claiming that diagnosis codes contain no information.

4.5 Medication features

The raw data has a separate column for each diabetes medication with values such as `No`, `Steady`, `Up`, and `Down`. The code counts how many drugs have an increased or decreased dosage and stores the result as `num_dosage_changes`. Most drug-specific columns are then removed. Insulin is retained as a categorical variable because it is a central diabetes treatment. The existing `change` and `diabetesMed` variables are converted to binary indicators.

This summary reduces dimensionality, but it loses which specific non-insulin drugs changed and whether the clinical patterns behind equal counts were different.

4.6 One-hot encoding, clipping, and scaling

Categorical features cannot be passed to logistic regression as arbitrary integer codes, because the integers would imply a false order. One-hot encoding creates one binary column per category. For example, insulin becomes four indicators for `No`, `Steady`, `Up`, and `Down`.

Selected count variables are transformed using fixed bounds:

Feature	Lower bound	Upper bound	Observed maximum
Time in hospital	1	14	14
Laboratory procedures	0	150	132
Procedures	0	10	6
Medications	0	100	81
Previous outpatient visits	0	50	42
Previous emergency visits	0	80	42
Previous inpatient visits	0	25	12
Dosage changes	0	24	within bound

Clipping and scaling are different operations. First,

$$x_{\text{clip}} = \min(\max(x, L), U),$$

so a value above U becomes U and a value below L becomes L. Second,

$$x_{\text{scaled}} = 2\frac{x_{\text{clip}} - L}{U - L} - 1,$$

which maps L to -1, the midpoint to 0, and U to 1. Fixed bounds give the same raw value the same meaning for all clients. In the current cohort, observed values are generally already inside the bounds, so the main effect is rescaling rather than truncating outliers.

Not every numerical column is scaled: `number_diagnoses` remains between 1 and 16. This is an implementation inconsistency and a limitation, particularly because DP gradient clipping is sensitive to feature scale.

The notebook performs one-hot encoding centrally before client partitioning. This guarantees a common feature schema, but it means the preprocessing stage itself is a simulation of FL rather than a decentralized preprocessing protocol. No target-dependent statistic is used in the fixed scaling, although the global category vocabulary is visible during encoding.

4.7 Final feature sets

The main cleaned file contains 135 features plus the target. An exploratory ablation also creates a 128-feature file that excludes medication-related features. The mean preliminary centralized results were:

Feature group	Features	AUROC	AUPRC
Core variables	8	0.594	0.133
Core + context, no discharge disposition	51	0.601	0.134
Core + context	72	0.636	0.161
Core + context + diagnoses	128	0.643	0.163
All features, including medication summaries	135	0.644	0.163

The 128-feature representation is saved, but `main.py` loads `data/preprocess/clean_data.csv`. Therefore, the reported federated experiments use all 135 features. The ablation is exploratory and must not be described as the feature-selection procedure used by the final FL runs.

5 Client partitioning

The source data does not contain a hospital identifier with which the original 130 institutions can be reconstructed. The project therefore simulates ten hospitals.

Client type	Assigned population	Training	Test
Client 0: small target hospital	500	400	100
Client 1: large hospital	25,000	20,000	5,000
Clients 2–8: background, each	about 5,433	4,346	1,087
Client 9: background	about 5,432	4,345	1,087
Client 0 holdout	1,000	—	1,000

Every client dataset is split 80/20, stratified by the target so that train and test prevalence remain similar. Client 0 also receives a separate 1,000-patient holdout. This is used because its ordinary test set has only about 100 patients and therefore yields unstable metrics for a rare outcome.

For the random setting, client 0 and its holdout pool are sampled randomly. For the non-IID setting, 80% of their combined 1,500-patient pool is selected from ages 70–99 and 20% from younger ages. The 500 client records and 1,000 holdout records are then taken from that shuffled pool. Client 1 is sampled next from the remaining population, and all remaining records are divided among the eight background clients.

The five seeds change the simulated client membership, train/test split, local shuffle order, and initial model weights. The partitions across seeds overlap because every seed resamples from the same original cohort; they are not five independent collections of hospitals.

6 Experimental methodology

6.1 Training configuration

The main model is a single-layer PyTorch logistic regression. The saved configuration uses:

Setting	Value
Clients	10
Federated rounds	5
Local epochs per round	1
Batch size	32
Optimizer	SGD
Learning rate	0.1
Client participation	all clients every round
Aggregation	sample-size-weighted FedAvg
Seeds	0, 1, 2, 3, 4
Input features	135

All clients train and evaluate in every round. The server also evaluates the global model on the union of all clients' test sets. Client 0 is additionally evaluated on its 1,000-record holdout.

6.2 Experimental arms

Arm	Meaning
Local-only	A scikit-learn logistic regression trained on client 0's 400 training records. It does not take part in FL.
Pooled	A centralized scikit-learn logistic regression trained on the concatenation of every client's training partition. It is an upper reference, not a privacy-preserving method.
Receive-only	Federated training without DP in which client 0 returns the unchanged model with zero examples. Clients 1–9 train normally. Client 0 receives and is evaluated on the global model.
Contribute, no DP	All ten clients participate in ordinary local SGD and FedAvg.
Contribute with DP	All ten participating clients use local DP-SGD with the same target (ε, δ) but client-specific noise multipliers.

The receive-only arm is essential for separating access to the federation from contribution. However, there is no *DP receive-only* arm. Consequently, comparing a DP arm with receive-only changes two things simultaneously: client 0 begins contributing, and every participating client changes from ordinary SGD to DP-SGD. That comparison cannot isolate the cost or benefit of client 0's contribution under DP.

6.3 DP configuration and accounting

Private runs use

$$\varepsilon \in \{1, 2, 4, 8, 16\}, \qquad \delta = 10^{-5}, \qquad C = 1.0.$$

Each client receives the same target ε and δ. Opacus calculates a separate noise multiplier σ from that client's sampling rate and total planned optimizer steps. With Poisson sampling, the code uses

$$q = \frac{1}{\text{number of local batches}}, \qquad T = 5 \times 1 \times \text{number of local batches}.$$

Because clients have different numbers of batches, equal target privacy budgets require different σ values.

Target ε	Small-client σ	Large-client σ	Final spent ε (approximately)
1	2.637	0.750	0.99–1.00
2	1.587	0.638	1.99
4	1.060	0.537	3.99–4.00
8	0.759	0.446	8.00
16	0.560	0.370	15.99–16.00

The small client needs a larger noise multiplier because its sampling rate is much higher: a batch represents a larger fraction of its local dataset. A separate PRV accountant is saved for each client and restored across federated rounds, so privacy loss composes over all five rounds. Logged final epsilon values are at or just below their targets.

The guarantee is record-level DP for each client's local DP-SGD process. It does not hide whether a hospital participated, and this implementation does not use secure aggregation. Opacus is configured with `secure_mode=False`, so its random-number generator is suitable for research simulation but not a cryptographically hardened deployment.

6.4 The MLP check

The proposal allowed a small multilayer perceptron (MLP) if logistic regression produced no detectable non-private MIA leakage. The tested MLP has 135 inputs, a 64-unit hidden ReLU layer, and one output, for 8,769 trainable parameters. It was tested on the random partition.

The saved MLP results contain only three arms: receive-only, non-private contribution, and DP at $\varepsilon = 8$. They do *not* contain the full epsilon sweep, and the saved MIA summary attacks only client 0. Any statement about MLP results at $\varepsilon = 1, 2, 4, 16$ or about its large-client MIA is not supported by the available output files.

6.5 Membership inference procedure

The final saved global model is attacked after round 5.

- For client 0, members are its 400 training records. Four hundred records are sampled without replacement from its 1,000-record holdout as non-members.
- For client 1, its 5,000 test records are the non-members. Five thousand records are sampled from its 20,000 training records as members.
- The same seed chooses the same patients across experimental arms, improving paired comparability.

Individual binary cross-entropy losses are calculated. The mean member loss is used as the classification threshold, while negative loss is used as the score for attack AUROC.

In receive-only, client 0's designated "member" records were actually excluded from training. That arm is therefore a useful negative control, but its client-0 member label should be understood as a candidate-member group rather than true membership.

6.6 Metrics

AUROC. The probability that a randomly chosen positive patient is ranked above a randomly chosen negative patient. It is insensitive to a fixed classification threshold. Chance is 0.5.

AUPRC. Summarizes precision and recall for the positive class. It is especially useful under class imbalance. A random ranking has expected AUPRC equal to prevalence: approximately 0.089 for the random client-0 holdout and 0.096 for the non-IID holdout.

Binary cross-entropy. Measures probabilistic prediction error; lower is better. The federated code records this as `loss`.

Brier score. Mean squared probability error. It appears in the preliminary centralized notebook, but it is not computed by the final federated evaluation code. It should not be reported as a federated result unless calculated separately.

MIA AUROC and accuracy. MIA AUROC is the primary threshold-free attack metric. Attack accuracy depends on the mean-member-loss threshold and is secondary.

6.7 Uncertainty

For a metric m measured on five seeds, the report uses

$$\bar{m} \pm t_{0.975,4} \frac{s}{\sqrt{5}}$$

as a 95% confidence interval for the mean, where $t_{0.975,4} \approx 2.776$. These intervals describe variability across the five simulated seeds. With only five seeds they are uncertain, and they do not represent variability across 130 real hospitals.

7 Implementation map

File	Responsibility
`preprocessing_and_baseline.ipynb`	Cleans the raw data, creates features, runs the preliminary centralized baseline and ablation, and writes the 135- and 128-feature CSV files.
`pipeline/partition.py`	Creates deterministic random or age-shifted client partitions and the client-0 holdout.
`federation/federation/task.py`	Defines the model, data loaders, local SGD training, and AUROC/AUPRC/loss evaluation.
`federation/federation/client_app.py`	Loads one client's data, performs ordinary or private local training, supports receive-only, evaluates locally, and logs metrics.
`federation/federation/server_app.py`	Initializes the global model, runs FedAvg, evaluates on the combined test set, and saves the final model.
`federation/federation/dp.py`	Calculates client-specific noise multipliers, attaches Opacus, and saves/restores the PRV accountant.
`pipeline/mia.py`	Runs the loss-based attack on the final model and saves patient-level scores and attack summaries.
`pipeline/baselines.py`	Fits local-only and pooled scikit-learn logistic-regression references on the same partitions.
`main.py`	Orchestrates partitions, five seeds, seven LR experimental arms, attacks, and baselines.
`analysis.ipynb`	Calculates confidence intervals, paired comparisons, tables, plots, and research-question summaries.

The simulation is implemented using Flower [8], PyTorch, scikit-learn, pandas, and Opacus.

8 Results

8.1 Small-hospital utility and MIA

Table 3 shows results for logistic regression with random client assignment. Values are mean ± 95% confidence interval over five seeds.

Table 3: Random partition, logistic regression, client-0 holdout and MIA.

Arm	Holdout AUROC	Holdout AUPRC	Small-client MIA AUROC
Local-only	0.539 ± 0.025	0.116 ± 0.017	—
Pooled	0.644 ± 0.015	0.190 ± 0.024	—
Receive-only	0.626 ± 0.026	0.148 ± 0.016	0.502 ± 0.022
Contribute, no DP	0.626 ± 0.026	0.148 ± 0.016	0.502 ± 0.022
DP, $\varepsilon = 1$	0.538 ± 0.050	0.106 ± 0.013	0.510 ± 0.040
DP, $\varepsilon = 2$	0.538 ± 0.036	0.106 ± 0.012	0.506 ± 0.041
DP, $\varepsilon = 4$	0.543 ± 0.047	0.106 ± 0.012	0.512 ± 0.042
DP, $\varepsilon = 8$	0.547 ± 0.050	0.107 ± 0.014	0.509 ± 0.043
DP, $\varepsilon = 16$	0.543 ± 0.042	0.106 ± 0.012	0.510 ± 0.043

Table 4 shows the age-shifted setting. These scores are evaluated on a different, older client-0 population and should not be read as a pure head-to-head test of IID versus non-IID difficulty.

Table 4: Age-shifted partition, logistic regression, client-0 holdout and MIA.

Arm	Holdout AUROC	Holdout AUPRC	Small-client MIA AUROC
Local-only	0.574 ± 0.031	0.139 ± 0.030	—
Pooled	0.647 ± 0.036	0.201 ± 0.021	—
Receive-only	0.639 ± 0.037	0.168 ± 0.021	0.499 ± 0.010
Contribute, no DP	0.639 ± 0.037	0.168 ± 0.021	0.499 ± 0.010
DP, $\varepsilon = 1$	0.554 ± 0.031	0.121 ± 0.014	0.509 ± 0.026
DP, $\varepsilon = 2$	0.567 ± 0.030	0.122 ± 0.014	0.510 ± 0.027
DP, $\varepsilon = 4$	0.562 ± 0.026	0.123 ± 0.012	0.509 ± 0.030
DP, $\varepsilon = 8$	0.559 ± 0.029	0.121 ± 0.011	0.509 ± 0.027
DP, $\varepsilon = 16$	0.561 ± 0.032	0.118 ± 0.010	0.509 ± 0.030

The saved MLP check is summarized in Table 5.

Table 5: Random partition, MLP check. Only $\varepsilon = 8$ was saved.

Arm	Holdout AUROC	Holdout AUPRC	Small-client MIA AUROC
Receive-only	0.633 ± 0.016	0.160 ± 0.018	0.504 ± 0.022
Contribute, no DP	0.633 ± 0.016	0.160 ± 0.020	0.504 ± 0.022
DP, $\varepsilon = 8$	0.524 ± 0.036	0.102 ± 0.014	0.508 ± 0.041

8.2 RQ1: at what epsilon does MIA fall to chance?

For the small hospital, logistic-regression MIA AUROC lies between 0.506 and 0.512 in every DP arm, and every across-seed 95% confidence interval includes 0.5. However, the attack is also at chance without DP: 0.502 ± 0.022 for the random partition and 0.499 ± 0.010 for the age-shifted partition. The MLP result is similar.

Answer to RQ1: no transition epsilon can be identified. The attack is at chance for every tested epsilon, but it is already at chance in the non-private baseline. The evidence therefore shows that this attack detected no membership signal; it does not show that a particular epsilon removed a previously detectable signal.

8.3 RQ2: does contributing beat receive-only?

Without DP, contribution and receive-only are nearly identical. Their paired per-seed AUROC difference is below 0.001. This is consistent with the small hospital carrying only 0.73% of the FedAvg weight.

Every DP arm is approximately 0.07–0.09 AUROC below the non-private receive-only model. This must not be interpreted as proof that contribution itself caused the drop. The DP arms apply DP-SGD to all clients, whereas receive-only uses ordinary SGD for clients 1–9. The comparison therefore combines the effect of client-0 contribution with the effect of changing the entire federation's optimizer to DP-SGD.

Answer to RQ2: non-private contribution provides no measurable advantage over receiving the model. The current design cannot isolate whether private contribution helps relative to a private receive-only federation because that control arm was not run.

8.4 RQ3: does local-only training beat the federation?

On the random partition, receive-only improves holdout AUROC from 0.539 to 0.626, a paired gain of about 0.087. On the age-shifted partition, it improves AUROC from 0.574 to 0.639, a gain of about 0.065. The non-private federated model is also within roughly 0.01–0.02 AUROC of the pooled centralized reference.

DP federated models are close to local-only in AUROC and generally below local-only in AUPRC. Their confidence intervals are broad, so the five-seed experiment does not demonstrate a reliable AUROC improvement over local-only training.

Answer to RQ3: local-only does not beat the non-private global model. Receiving the non-private federation is clearly more useful for client 0. Under the tested DP setup, neither a reliable advantage nor a reliable disadvantage over local-only AUROC is established, although AUPRC is lower.

8.5 RQ4: small versus large hospital

The large-hospital LR attack is also close to chance: approximately 0.494–0.495 in the random partition and 0.500–0.504 in the age-shifted partition. Its intervals are narrower because the attack compares 5,000 members with 5,000 non-members, rather than 400 with 400 for the small hospital.

For utility, the large hospital's non-private test AUROC is approximately 0.620 in the random partition and 0.625 in the age-shifted partition. Under DP it falls to approximately 0.541–0.549 and 0.526–0.530, respectively. Both small- and large-hospital utility decline by roughly 0.07–0.10 compared with their corresponding non-private models.

Answer to RQ4: the implementation gives both hospitals the same target epsilon but uses different noise multipliers, with substantially more noise at the small client. Nevertheless, the chosen MIA is at chance for both hospitals, so it cannot demonstrate a difference in empirical attack protection. Both hospitals show a substantial utility drop under the tested DP training schedule.

8.6 Why increasing epsilon did not clearly improve utility

Normally, increasing ε lowers the required noise multiplier, so one might expect utility to improve. That monotonic pattern is not clear here. Several factors may explain the observation:

- training lasts only five rounds with one local epoch per round;
- the global DP curves are still changing at round 5, suggesting incomplete convergence;
- clipping can introduce bias even when less noise is added;
- random variation across only five seeds is large relative to the small epsilon differences; and
- client heterogeneity and weighted aggregation mix gradients with different noise multipliers.

These are plausible explanations, not separately tested causal conclusions. The safe conclusion is simply that the experiment found no consistent utility improvement across $\varepsilon \in \{1, 2, 4, 8, 16\}$ under the fixed five-round schedule.

9 Discussion

9.1 What the project demonstrates

The strongest result is a utility result, not an attack result. A model learned from the larger federation generalizes better to the small hospital's holdout than a logistic regression trained on only 400 local records. Client 0 does not need to contribute its own records to receive nearly all of that benefit because its potential contribution is tiny compared with the other 54,767 training records.

The pooled model remains slightly better than the federated model. This is reasonable: pooled scikit-learn logistic regression directly optimizes over all training rows until convergence, whereas the FL model uses only five rounds of distributed SGD. The two procedures are not optimization-identical, so the pooled model is a reference rather than a perfectly matched control.

The tested DP configuration achieves the intended accounting targets but substantially reduces predictive utility. This does not imply that DP-SGD is generally unsuitable. It shows that this particular combination of five rounds, one local epoch, learning rate 0.1, clipping norm 1.0, Poisson sampling, and client-specific noise did not train a competitive model. More extensive tuning would itself need to respect a clean evaluation protocol.

9.2 What the project does not demonstrate

An attack AUROC near 0.5 is not proof that the model leaks no private information. It means this particular loss-based attack, with this knowledge and these samples, could not distinguish members from non-members. Stronger attacks could use prediction vectors, gradients, shadow models, label-specific thresholds, or other auxiliary information.

Similarly, the absence of a DP utility trend across epsilon does not mean epsilon is irrelevant. The privacy accountant clearly changes sigma with epsilon. The observed utility differences may be hidden by undertraining and seed variability.

Finally, the age-shifted result is not a direct test on real hospital heterogeneity. It is a controlled synthetic partition based only on age, constructed after centralized preprocessing.

9.3 Recommended decision for the simulated small hospital

If the available arms are treated literally, receive-only gives the small hospital the best observed balance: it substantially outperforms local-only training, matches non-private contribution, and does not use client 0's records for training. This is an empirical conclusion for the simulation, not a general policy recommendation. In a real deployment, access rules, fairness, governance, distribution changes, and whether other hospitals require reciprocal contribution would also matter.

There is not enough evidence to recommend one tested epsilon as the "best" privacy–utility point. All DP arms have similar, lower utility, and the attack is already at chance without DP.

10 Limitations and threats to validity

1. **Synthetic hospitals.** The original institution labels are unavailable, so client partitions do not reproduce real hospital populations, workflows, or correlations.
2. **Central preprocessing.** One-hot encoding and feature construction occur before federation. The experiment evaluates federated training, not a completely decentralized data-engineering pipeline.

3. **Attack weakness.** The loss-based MIA is at chance without DP. This creates a floor effect and prevents a convincing measurement of attack-risk reduction.
4. **No DP receive-only control.** The DP-versus-receive-only comparison changes both client participation and the training algorithm used by the other clients.
5. **Short training.** Five rounds may be insufficient for DP-SGD convergence. The conclusions apply to the tested schedule.
6. **Only five seeds.** Confidence intervals are based on $n = 5$ and are therefore wide and sensitive to individual runs.
7. **Overlapping simulated populations.** Partitions are regenerated from the same cohort for each seed; seeds are not independent real-world hospitals.
8. **Repeated holdout evaluation.** Client 0's holdout is evaluated after every round and used in analysis. If it influenced tuning or experiment selection, it no longer functions as a completely untouched final test set.
9. **Feature-selection mismatch.** The notebook saves a 128-feature alternative, but the FL pipeline uses the 135-feature file.
10. **Scaling inconsistency.** `number_diagnoses` is not scaled while selected count variables are. Feature magnitude can affect SGD and DP clipping.
11. **Demographic analysis unavailable.** Race and gender are removed, so subgroup fairness and privacy cannot be assessed. Removing them does not remove proxy information.
12. **Prediction time.** Discharge disposition is retained. This is valid only because the intended prediction point is discharge; it would be future-information leakage for a model intended for use at admission.
13. **Research-mode randomness.** Opacus uses `secure_mode=False`. The noise mechanism is suitable for experiments but lacks the cryptographically secure random-number generation recommended for deployment.
14. **No secure aggregation.** FL reduces direct record sharing but the server receives individual model updates. DP limits record influence; it does not provide all system-level protections required by a real medical deployment.
15. **Model scope.** Logistic regression and one small MLP cover only a narrow range of model capacity. The MLP output files support only $\varepsilon = 8$ and only a small-client MIA.

11 Conclusion

This project evaluates how a simulated small hospital fares under local training, non-private federated learning, and client-side DP-SGD. The small hospital benefits substantially from receiving a model trained by the federation: non-private global AUROC is about 0.63–0.64, compared with about 0.54–0.57 for local-only logistic regression. Contributing its own 400 training records adds almost no measurable utility because those records make up only 0.73% of the FedAvg weight.

Under the tested DP configuration, utility falls to approximately 0.54–0.57 and does not improve consistently as epsilon increases. Accounting behaves as intended, with each client finishing near its target epsilon and the small client requiring a larger noise multiplier. However, the loss-based membership attack is already at chance without DP. The study therefore cannot claim that DP caused MIA AUROC to fall to chance or select an optimal epsilon from attack performance.

The defensible overall conclusion is that the simulated small hospital gains predictive utility from access to a non-private federation, but the tested private training schedule does not preserve that gain. A stronger study would add a DP receive-only control, reserve an untouched final test set, train DP models to convergence, and evaluate multiple membership attacks before drawing a privacy–utility recommendation.

A How to explain the project orally

Why keep only one encounter per patient?

If the same patient appeared in training and testing through different admissions, the evaluation could reward patient-specific recognition rather than generalization to new patients. Keeping the earliest encounter makes each row represent a different patient.

Why group ICD-9 codes?

The codes are categories, not measurements. Exact one-hot encoding would create many rare columns. Broad disease groups provide better-supported indicators but sacrifice fine diagnostic detail.

Why clip and scale?

Clipping caps extreme values; scaling maps different count variables to comparable numerical ranges. Fixed bounds give the same raw value the same representation everywhere. In this dataset the bounds mostly act as a scaling definition because observed values already fall within them.

Why use AUPRC as well as AUROC?

Only about 9% of patients are readmitted within 30 days. AUROC can look acceptable even when positive predictions are weak. AUPRC focuses on performance for the rare positive class, and its random baseline equals the positive prevalence.

Why are receive-only and contribution almost identical?

FedAvg weights by training size. Client 0 has 400 of 55,167 training records, so its update receives only 0.73% of the aggregate weight.

Did DP stop the membership attack?

We cannot show that. The attack was already at chance without DP. We can say that no tested model showed detectable leakage under this attack, but not that there was no leakage or that DP caused the result.

Why did larger epsilon not clearly help?

Larger epsilon did reduce the noise multiplier, but five rounds may not have been enough for stable convergence, and variability over five seeds was large. The experiment observed no monotonic utility trend; it did not prove that epsilon has no effect.

What is the main takeaway?

The small hospital benefits from receiving the global model. Its own data adds little because it is a tiny fraction of the federation. The tested DP configuration protects each local training process according to the accountant but loses much of the global model's utility.

B Reproducibility checklist

- Raw data: Diabetes 130-US Hospitals for Years 1999–2008.
- Final cohort: 69,963 unique patients; 6,275 positives.
- Final FL input: 135 features plus target.
- Seeds: 0–4.

- Main runs: random LR and age-shifted LR.
- Experimental arms: receive-only, non-private, and DP at $\varepsilon = 1, 2, 4, 8, 16$.
- DP constants: $\delta = 10^{-5}$ and clipping norm 1.0.
- Training: five rounds, one local epoch, batch size 32, SGD learning rate 0.1.
- Primary utility metrics: holdout AUROC and AUPRC.
- Primary privacy-risk metric: loss-based MIA AUROC.
- Statistical summary: mean and t-based 95% CI across five seeds.

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